Faculty of Engineering

Module Title Vibration and Waves

Revision 1&2&3 [Answered]

(Q1) Choose the correct answer:

(1) Physical quantities have only magnitude and no specific direction

(a) Scalar Quantities (b) Derived Quantities (c) Vector Quantities

(2) Physical quantities have both magnitude and direction

(a) Scalar Quantities (b) Derived Quantities (c) Vector Quantities

(3)The S.I unit of energy is ……

A) watt B) newton C) joule D) second

(4) The S.I unit of force is ……

A) watt B) newton C) joule D) second

(3)…….. is motion of an object that regularly returns to a given position after a fixed time interval

(a) Amplitude (b) Direct motion (c)Periodic motion

(4)…….. oscillatory motion where an object moves back and forth around an equilibrium (central) position

(a) Direct motion (b) Simple harmonic motion (c) Frequency

(5) ……the maximum displacement in direction of vibration.

a) Amplitude b) Frequency c) Wavelength

(6)…….. is number of complete cycles per unit of time

a) Amplitude b) Frequency c) Wavelength

(7) ……is time needed to make one cycle

a) Amplitude b) Frequency c) Periodic Time

(8) **.....**is a disturbance that transfers energy through space or a medium without transporting matter

a) wave b) Frequency c) Wavelength

(9) .....is the distance between two successive points in the wave that have the same phase.

a) wave b) Frequency c) Wavelength

(10) .....is the phenomenon that occurs when two or more waves meet and overlap in the same medium

(a) Periodic Time (b) Wave interference (c) Wavelength

(11..... is the Interference of two identical sinusoidal waves moving in opposite directions.

a) Wavelength b) Periodic Time c) standing wave.

(12) .....is the positions of zero amplitude in standing wave eq.

a) Antinodes b) Nodes c) Wavelength

(13) .....is the positions of maximum amplitude in standing wave eq
a) Wavelength b) Antinodes c) Nodes

(Q2) Classify the following as fundamental or derived quantities:

(a)Length (b) Mass (c)Velocity (d)Force

Answer:

(a)Length is fundamental

(b) Mass is fundamental

(c)Velocity is derived

(d)Force is derived

(Q3) State whether each quantity is scalar or vector:

(a)Time (b) Displacement (c)Speed (d)Acceleration

Answer:

(a)Time is scalar

(b) Displacement is vector

(c)Speed is scalar

(d)Acceleration is vector

(e) Force is vector

(Q4)Problems

(1) Using the method of dimensions ,check the validity of:

$$x - x_{0=} v_0 t + \frac{1}{2} vt^2$$

Answer:

i) **For L.H.S** **X - X_0 = L**

ii) **For R.H.S** $v_0\, t + \frac{1}{2} vt^2 = L + LT$

So the relation is not correct

(2) Find the relation between the velocities of transverse waves produced from the vibration Of thin homogeneous string and between the tension in the string and unit length of it.

Answer:

$v = L\,T^{-1}$

$T = ML\,T^{-2}$

$\mu = M\,L^{-1}$

$v = k\,T^a \mu^b$

$L\,T^{-1} = [ML\,T^{-2}]^a + [M\,L^{-1}]^b$

For T $-1 = -2a$ $a = \frac{1}{2}$

For L 1=a+-b b=-1/2

$v = k\,T^{1/2} \mu^{-1/2}$

$$v = k \sqrt{\frac{T}{\mu}}$$

(3) An object oscillates with simple harmonic motion along the x axis. Its displacement from the origin varies with time according to the equation:

$$X = (4.00\ m)\cos(\pi t + \pi/4)$$

where t is in seconds and the angles in the parentheses are in radians.

(a) Determine the amplitude, frequency, and period of the motion. (b) Determine the maximum speed and maximum acceleration of the object.

Answer:

$$A = 4.00\ m$$

$$\because \omega t = \pi t \ \therefore\ \omega = \pi \ \rightarrow\ \pi = 2\pi f \qquad \therefore f = 0.5\ Hz$$

$$T = \frac{1}{f} = \frac{1}{(0.5)} = 2\ sec$$

$$v_{max} = \omega A = (\pi)(4\ m) = 12.6\ ms^{-1}$$

$$a_{max} = \omega^2 A = (\pi)^2(4\ m) = 39.5\ ms^{-2}$$

(4)A simple harmonic oscillator takes 20 s to undergo four complete vibrations. Find

(a) the period of its motion, (b) the frequency in hertz, and (c) the angular frequency in radians per second.

Solution:

$T = \frac{total\ time}{number\ of\ oscillations} = \frac{20}{4} = 5$ s

f= $\frac{1}{T} = 0.2HZ$

ω = $\frac{2\pi}{T}$ =1.26 rad/s

(5)A spring has a spring constant of 100N/m and the period of oscillation is 0.628 s .Find the mass attached to the spring

Solution:

$T = 2\pi\sqrt{\frac{m}{k}}$ $\qquad$ $m = \sqrt{\frac{T^2\,K}{4m^2}} = 1\ kg$

(6) **A block with a mass of 200 g is connected to a light spring for which the force constant is 5 N/m and is free to oscillate on a horizontal, frictionless surface. The block is displaced 5 cm from equilibrium and released from rest (a) Find the period of its motion. (b) Determine the maximum speed of the block. (c) What is the maximum acceleration of the block?**

Solution:

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{(200\times10^{-3}\ kg)}{(5\ Nm^{-1})}} = 1.26\ s$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{(1.26\ s)} = 5\ rad/s$$

$$v_{max} = \omega A = \left(5\frac{rad}{s}\right)(5\times10^{-2}\ m) = 0.25\ ms^{-1}$$

$$a_{max} = \omega^2 A = \left(5\frac{rad}{s}\right)^2(5\times10^{-2}\ m) = 1.25\ ms^{-2}$$

(7) A simple pendulum has a time period of 3 s on Earth. Calculate its length taking g= 9.8 m/ s^2

Solution:

$T = 2\pi\sqrt{\frac{L}{g}}$ $\qquad$ 3= $2\pi\sqrt{\frac{L}{9.8}}$ $\qquad$ L = $2.234\ m$

(8) A simple pendulum makes 120 oscillations in 4 minutes. Find

(a)Its periodic time (b) Its frequency (c) Its length ($g = 9.8\frac{m}{s^2}$)

Solution

$T = \frac{total\ time}{number\ of\ oscillations}$ = $\frac{4*60}{120}$= 20 s

f= $\frac{1}{T} = 0.05\ HZ$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$L = \frac{gT^2}{4\pi^2} = 99.294\ m$$

(9) A string is under a tension of 200N .Its linear mass density is 0.02 kg/m .Find the speed of the wave

Solution:

$v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{200}{0.02}}$ = 100m/s

(10) A string of length 40m has mass 1.2kg and tension 750N .A sinusoidal wave on it has amplitude 3 cm and frequency 25HZ.

Find: linear density , wave speed, angular frequency and average power.

Solution:

$\mu = \frac{m}{l} = \frac{1.2}{40} = 0.03kg/m$

$v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{750}{0.03}}$ = 158.11m/s

$$\omega = 2\pi f = 157.08\ rad/s$$

$P_{av} = \frac{1}{2}\ \mu\omega^2 A^2 V$ = 52.67 W

(11)

A string fixed at both ends is 8.4 m long and has a mass of 0.12 Kg; it is subjected to a tension of 96 N and starts to oscillate. (a) What is the speed of the waves on the string? (b) What is the longest possible wavelength for the standing waves? (c) Give the frequency of the wave on the string.

$$\because \mu = \frac{m}{L} = \frac{0.12}{8.4} = 0.0143\ kg/m \qquad ,\because v = \sqrt{\frac{\tau}{\mu}} \qquad \therefore v = \sqrt{\frac{96}{0.0143}} = 81.93\ m/s$$

The longest wavelength (when it makes one loop, n=1)

$$,\because \lambda = \frac{2L}{n} \qquad \therefore \lambda = \frac{2 \times 8.4}{1} = 16.8\ m$$

$$,\because f = \frac{v}{\lambda} \qquad \therefore f = \frac{81.93}{16.8} = 4.88\ Hz$$